

Estimating spatial firm productivity spillovers using UK microdata

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DATA

I have used data from Fame for this investigation. Fame is a comprehensive set of financial datapoints from UK firms, with instruments appropriate for this analysis, in particular number of employees and balance sheet data from which economically interesting variables can be derived. Fame is provided by Moody's Analytics, an international credit agency, and collects its data from UK firms via survey³.

Its international counterpart, Orbis, offers a similar service for international counterparts. This analysis could therefore be extended to offer insights comparing the spillover effects of UK firms to their international counterparts. For this investigation, the UK dataset is comprehensive and the most policy-relevant for this investigation.

Raw data was accessed from Fame in February 2025 by Professor Lars Nesheim at UCL for (Irastorza-Fadrique et al., 2026). This project stored a series of Excel tables splitting financial balance sheet variables by their category. Downloadable indicators under 'a1_ID' and 'a5_misc' contain fixed datapoints about each firm; 'a2_key_finance', 'a3_assets', and 'a4_profits' contain yearly financial data about each firm. I designed a series of Python scripts to ingest this data and store it in a 'DuckDB' database, separated into fixed and yearly datapoint tables. 'DuckDB' is an open-source, local, relational (SQL) architecture, designed for processing bulk queries across columns of variables, making it well-suited for econometric data analysis.

³Moody's Analytics (2026)

MODEL

I start by estimating a basic 2-factor Solow production model at the firm level.

Labour input is taken from the number of employees reported on the FAME balance sheet per firm. This dataset, like all others, does not include hours worked per employee on a firm-level that would offer a more granular productivity (GVA per hour worked) measures. An estimation in this setting would likely have to impute the measure from industry aggregates, which would likely confound the true firm-level signal.

Capital is FAME's `fixed_total` variable, which represents the sum of tangibles, intangibles, and other investments within the balance sheet, namely an extremely comprehensive measure of the firm's fixed capital. The average capital-output ratio is 6x within FAME when filtering for positive values, which is moderately higher than macroeconomic aggregates typically suggest likely due to accounting standard differences.

TABLE I
Capital-output ratios in FAME corpus of firms

	tangibles	fixed_other (tangibles + intangibles + inv._other)	total_assets (fixed_total + current_assets)
Capital-output ratio	3.53	6.00	11.56

SPATIAL ANALYSIS

TABLE II
Baseline TFP calculation from 2-factor production

Parameter	gva1_ft	gva2_ft	k2_ft	gva1_f	gva1_t	gva1_
β_0	3.201	3.445	4.840	3.223	1.994	3.201
const	(0.032)***	(0.051)***	(0.019)***	(0.027)***	(0.010)***	(0.032)***
β_1	0.621	0.624	0.717	0.618	0.559	0.621
ln_employees	(0.005)***	(0.007)***	(0.005)***	(0.004)***	(0.002)***	(0.005)***
β_2			0.058			
ln_fixed_total			(0.001)***			
β_3	0.265	0.245		0.264	0.422	0.265
ln_total_assets	(0.004)***	(0.007)***		(0.004)***	(0.002)***	(0.004)***
Observations	1,081,520	618,588	1,042,854	1,081,520	1,081,520	1,081,520
i	140,458	93,041	135,428	140,458		140,458
t	19	19	19		19	19
$R^2(\text{within})$	0.386	0.323	0.339	0.386	0.366	0.386

TABLE III
Static panel linear-in-means using spatial-group TFP averages

Parameter	lmm	lmm_no_firm_fe	lmm_employees
β_0	2.075	0.310	1.202
const	(0.044)***	(0.038)***	(0.215)***
β_1	0.033	0.204	0.306
peer_tfp_ttwa_donut	(0.013)***	(0.013)***	(0.071)***
β_2	0.015	0.142	0.015
peer_tfp_pc4_donut	(0.005)***	(0.008)***	(0.005)***
β_3	0.280	0.559	0.280
peer_tfp_pc8	(0.005)***	(0.004)***	(0.005)***
β_4			-0.058
peer_tfp_pc8_x_employees			(0.015)***
β_5			0.184
ln_employees			(0.044)***
Observations	764,537	764,537	764,537
i	107,143		107,143
t	19	19	19
$R^2(within)$	0.040	-0.000	0.040

TABLE IV
Static panel using continuous distance-decay weighted TFP average of peers

Parameter	dd_gravity1	dd_neg2	dd_expdd3	dd_struct1	dd_struct2	dd_struct3
β_0	2.479	2.586	2.417	2.698	2.799	2.652
const	(0.029)***	(0.029)***	(0.032)***	(0.033)***	(0.032)***	(0.035)***
β_1	0.168			0.168		
tfp_wav1	(0.003)***			(0.003)***		
β_2		0.135			0.135	
tfp_wav2		(0.003)***			(0.003)***	
β_3			0.184			0.183
tfp_wav3			(0.005)***			(0.005)***
β_4	-0.009	-0.009	-0.009	0.619	0.619	0.619
ln_employees	(0.004)**	(0.004)**	(0.004)**	(0.005)***	(0.005)***	(0.005)***
β_5	0.013	0.012	0.016			
ln_nb_peers	(0.004)***	(0.004)***	(0.004)***			
β_6				0.265	0.265	0.265
ln_total_assets				(0.004)***	(0.004)***	(0.004)***
Observations	1,067,013	1,067,013	1,067,013	1,067,013	1,067,013	1,067,013
i	139,162	139,162	139,162	139,162	139,162	139,162
t	19	19	19	19	19	19
$R^2(within)$	0.010	0.010	0.004	0.392	0.391	0.388

TABLE V
Dynamic panel using instrumented continuous distance-decay weighted TFP

	(1) $\ln(gva_1)$	(2) $\ln(gva_1)$	(3) $\ln(gva_1)$
constant	1.371 (0.134)***	1.577 (0.144)***	1.064 (0.103)***
$\ln(gva_1)_{t-1}$	0.005 (0.001)***	0.005 (0.001)***	0.005 (0.001)***
$\ln(employees)$	0.696 (0.038)***	0.689 (0.037)***	0.709 (0.040)***
$\ln(total_assets)$	0.259 (0.041)***	0.266 (0.041)***	0.247 (0.044)***
tfp_wav_1	0.503 (0.028)***		
tfp_wav_2		0.423 (0.023)***	
tfp_wav_3			0.618 (0.042)***
Hansen Test	X (0.417)	X (0.459)	X (0.246)
AR(1) Test	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***
AR(2) Test	X (0.746)	X (0.600)	X (0.945)
i	139,162	139,162	139,162
Observations	713,339	713,339	713,339

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APPENDIX 1

APPENDIX 2

APPENDIX 3